PROJECT

LOW COST SOLAR-TRACKING WITH

PMDC MOTORS

Project by :-

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Problem statement:-

Solar panels generate maximum power only when directly facing the sun, but fixed installations lose efficiency as the sun moves. Solar tracking systems solve this by continuously aligning panels with sunlight, boosting energy output by up to 40%.

Traditional trackers often use stepper or shunt motors, which require complex control and added cost. For rural, low-cost applications, this complexity is impractical. Permanent Magnet DC (PMDC) motors provide a simpler alternative: their speed is controlled directly by armature voltage (via PWM), and direction by polarity reversal through an H-Bridge.

This project addresses the challenge of designing a low-cost, sensor-based solar tracker using PMDC motors, LDR sensors, and microcontroller control, to maximize solar panel efficiency in a practical and sustainable way.

Basic Idea of Project :-

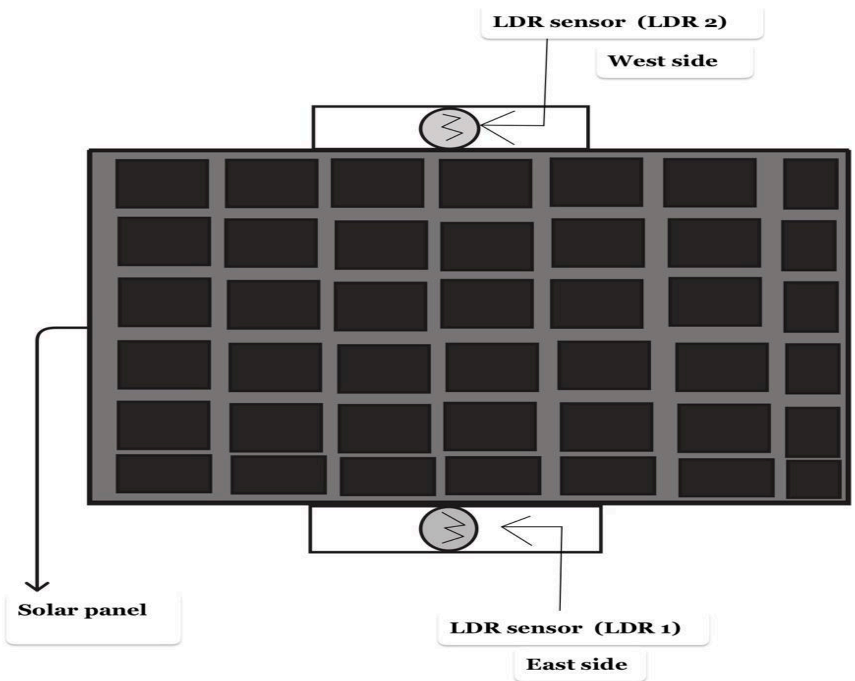
In the solar tracking system, two Light Dependent Resistors (LDRs) are placed at opposite edges of the solar panel to sense light intensity. When both LDRs receive equal illumination, their output voltages are equal, resulting in no error signal to the controller. Consequently, no control signal is applied to the H-bridge, and the PMDC motor remains stationary.

However, when one LDR receives higher light intensity than the other, a difference in their output voltages is generated. This difference is processed by the controller, which produces appropriate control signals for the H-bridge and PWM . The H-bridge then drives the PMDC motor in a specific direction by controlling the current flow through it,while the PWM controls the speed of the motor.

As a result, the motor rotates the solar panel toward the direction of higher light intensity until both LDRs receive the required results. At this point, the motor stops, ensuring that the panel is aligned for maximum solar energy capture.

If both LDRs receive very low light (darkness), the controller detects this condition and sends a reverse signal to the H-bridge, causing the PMDC motor to rotate the panel back to its initial position. However, an additional mechanism such as a limit switch or timing control is required to stop the motor at the desired position.

THE SOLAR PANEL STRUCTURE DIAGRAM WITH LDR SENSOR:-



Systematic diagram of LDR sensors mounted on solar panel

THE SPEED CONTROL OF PMDC MOTORS:-

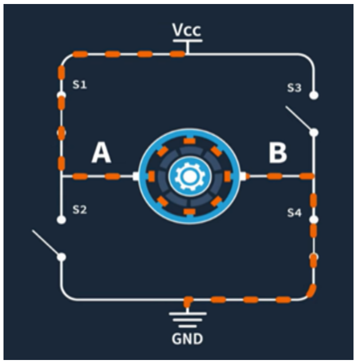
Permanent Magnet DC (PMDC) motors efficiency is higher than wound-field DC motors since permanent magnets provide the field flux without requiring extra field windings, which eliminates copper losses and reduces power consumption. The speed of a PMDC motor is directly proportional to the average armature voltage, meaning that by controlling the voltage supplied, the motor speed can be finely adjusted.

N ɑ V - - - - - -(speed is directly proportional to input voltage)

PWM, or Pulse Width Modulation, is basically a way of controlling a DC motor's speed by turning the supply on and off very quickly at a fixed frequency, say 1 kHz. Each cycle has an ON time (Ton) and an OFF time (Toff), and the ratio of Ton to the total cycle time is called the duty cycle. **Example-If the duty cycle is 100%, the motor gets the full supply voltage and runs at maximum speed. At 50%, the motor only sees half the average voltage and runs at about half speed. At 0%, no voltage is applied and the motor stops**.

Even though the motor is technically receiving pulses, its inductance and mechanical inertia smooth out the effect, so it doesn't jerk with each pulse. Instead, it responds to the **average voltage across the armature**. That's why PWM is such an effective way of simulating a variable DC voltage using a fixed supply, giving you precise and efficient control of motor speed without wasting energy in resistors or mechanical methods.

H-Bridge Bi-Directional Control:-



In solar tracker:
- Panel must move left (morning) and right (evening)

So bidirectional rotation is required
- Therefore → H-Bridge is needed

How H-Bridge Controls Direction of PMDC Motor
H-bridge controls direction by reversing current through the motor

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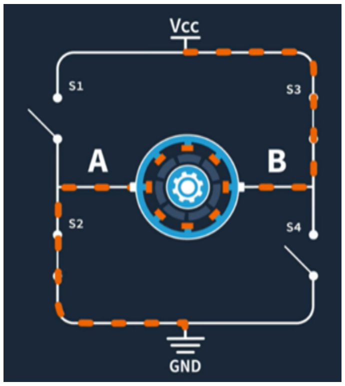
Case 1: Forward Rotation
- Switches ON: S1 and S4

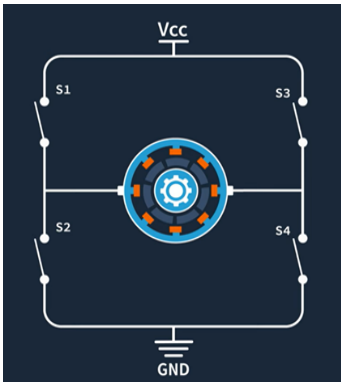
 # Current path:
Vcc → S1 → Motor → S4 → GND
Motor rotates forward



Case 2: Reverse Rotation
- Switches ON: S3 and S2

#Current path:
Vcc → S3 → Motor → S2 → GND
Current direction reverses
Motor rotates reverse

Case 3: No Rotation (All Switches OFF)
• Switches ON: None (S1, S2, S3, S4 are OFF)
 # Current path:
No closed path exists → No current flows through the motor



Working:-

Step1:- Sunlight Detection (Sensors)

Two LDR sensors are mounted on either side of the solar panel with the panel in the middle. This arrangement helps detect the difference in light intensity and identify the sun's position. When one LDR receives more sunlight than the other, it indicates that the sun has shifted. The LDRs change their resistance depending on the light falling on them. This resistance is converted into a voltage signal using a voltage divider circuit. Using amplifiers , the voltage signals are processed and sent to the microcontroller for further decision making.

Step2:- Signal Processing (Controller)

The microcontroller (Arduino) reads the voltage values from both LDRs. It compares them to determine which side has higher intensity. If the upper LDR shows more light, the controller decides to rotate the panel upper; if the lower LDR shows more light, it rotates lower. When both readings are equal,or the difference is less than equal to a particular threshold than panel is aligned and no movement is required.This difference in the voltage will be used in feedback and H bridge control

Step3:-Closed-Loop PWM Control

As already known the PMDC motors can be controlled with the PWM generator.
Based on the difference in sensor readings:

- **Large difference** → Arduino sets a higher PWM duty cycle → (motor runs faster to quickly correct alignment.)

- **Small difference** → Arduino sets a lower PWM duty cycle → (motor runs slower for fine adjustment.)
- This ensures smooth operation and prevents overshooting.

If an **external PWM generator (555/LM430)** is used:

- Arduino sends a control signal to adjust the duty cycle of the PWM generator.
- The PWM waveform is then fed into the **Enable pin of the H-Bridge**.

Step4:- Motor Driver (H-Bridge)

Arduino simultaneously sends direction signals (HIGH/LOW) to the H-Bridge input pins. The H-Bridge combines the PWM waveform (speed control) and the Arduino's direction signals.

- Input pins decide the polarity → motor direction.
- Enable pin receives PWM → motor speed.
  The driver amplifies these signals to supply the required current and voltage to the PMDC motor.

Step 5 – Panel Actuation (PMDC Motor)

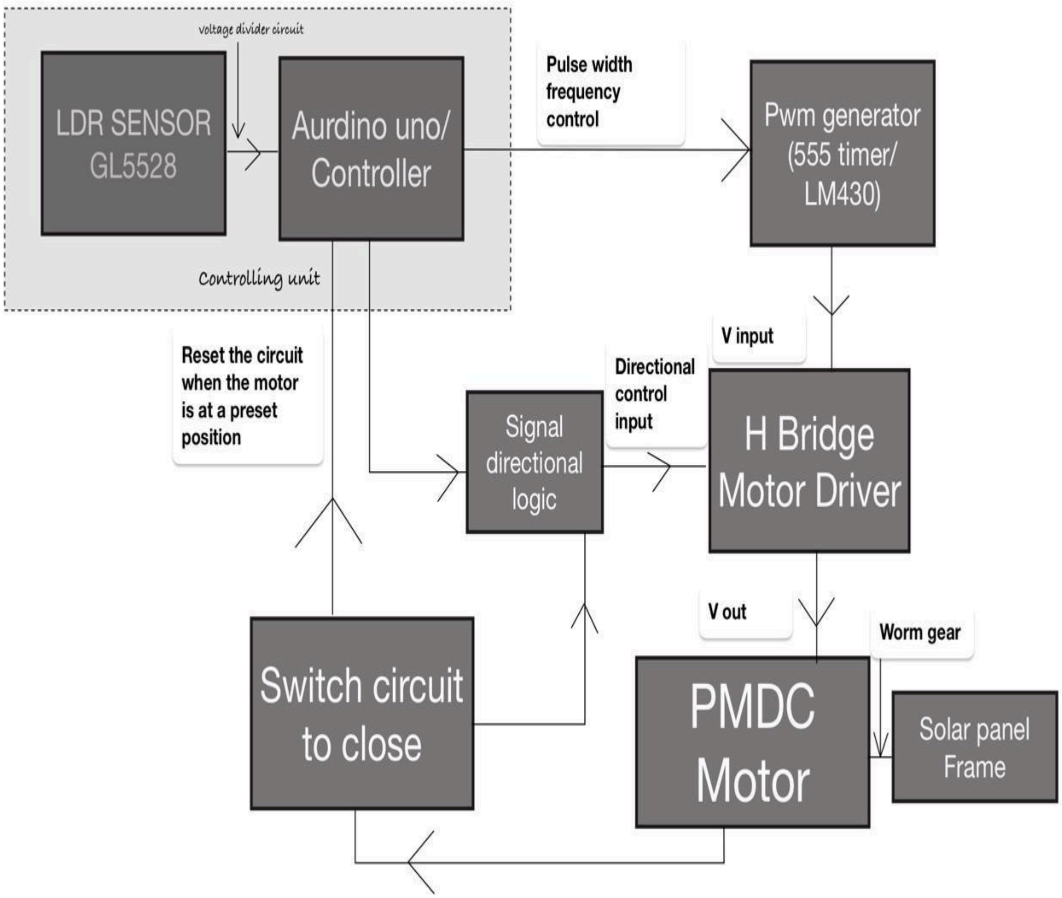
The PMDC motor rotates the solar panel along its axis until both LDRs detect intensity as per our required (The voltage difference received in arduino as signal less then a threshold) . At that point, Arduino reduces PWM duty cycle to zero resulting in the motor stops. This cycle repeats throughout the day, keeping the panel aligned with the sun.

Step 6 – Return to Origin (Low-Intensity Condition)

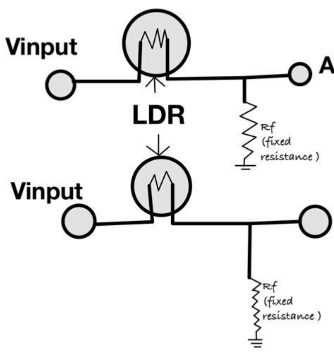
If the **overall light intensity remains below a threshold for a set duration** (for example, evening or cloudy conditions):

- Arduino detects that both LDR voltages are consistently low.
- A timer or counter in the program confirms that this condition persists for a defined period.
- Arduino then sends a command to the H-Bridge to rotate the motor in the **reverse direction** until the panel reaches its **origin/home position** (usually east).
- A **limit switch or sensor** at the origin detects when the panel is back in place, and the motor is stopped.
  This ensures the panel resets daily and is ready to track the sun again the next morning.

BLOCK DIAGRAM



voltage divider circuit



$$A_0 = V_{in}\frac{R_F}{(R_F + R_0)}$$

$$A_1 = v_{in}\frac{R_F}{(R_F + R_1)}$$

A1 and AO are the sensory input of LDR to auridino UNO

CONCLUSION:-

The static limitation of conventional solar panels is overcome by employing a closed-loop solar tracking system that continuously aligns the panel with the sun's movement from east to west. Using LDR sensors for real-time monitoring, the Arduino controller dynamically adjusts the PMDC motor through PWM speed control and H-Bridge directional logic. This ensures precise orientation of the panel throughout the day, preventing energy loss due to misalignment. As a result, the solar panel operates closer to its maximum power point, significantly improving energy conversion efficiency. Compared to fixed installations, such tracking systems can enhance overall output by 25–40%, making them a cost-effective and efficient solution for renewable energy applications.

THANK YOU